https://pigeonhole.at/1280836

Project Work

CSD 3156 Part II Cloud Computing

Project Requirements

- **Objectives**

  The project for CSD 3156 Part 2 Cloud Computing is to assess the students' knowledge of cloud computing and ability to develop cloud application services by utilising various cloud technologies and services from public clouds, e.g., AWS, GCP, Azure, etc.

- **Requirements**

  - Students are required to develop a cloud-based web application which can interact with users to provide information or complete certain tasks over the Internet.
  - The application shall be functional (**), scalable (*), reliable (*), elastic (*) and secure (*). The number of stars indicates the priority of each feature in this work.
  - Students can work in a group (≤6 students). Submit a proposal and indicate the team members at https://bit.ly/3vwA1oZ **by the end of Week 11.**
  - Students shall submit a report, a video demo and source code **by the end of Week 13.**

Submission Materials

- ✓ **Project proposal (5 marks)**
  - Submit to x-site by Week 11 (22 Mar 2026, Sunday, 11:59 PM)
  - Indicating the overall design, schedule, and team members and their roles
  - No more than **4 pages** (excluding the cover page) according to the provided template.
- ✓ **Project report**, **demo video**, and **source code** (**20 marks**)
  - Submit to x-site by Week 13 (5 Apr 2026, Sunday, 11:59 PM)
  - **Report:** Please provide details on how it functions including features like scalability, reliability, and security within **6 pages** (excluding cover page and references) using a provided template.
  - Team must include the percentage (%) contribution of each team member in the project report, and the total contribution of all the team members shall be 100%.
  - **Video:** Showcase your team's efforts by giving a demonstration that clearly explains the various functions and features.
  - **Source code:** Submit complete source code the x-site dropbox.
- ✓ **Individual contribution (5 marks)**
  - Submit the project report by Week 13 (5 Apr 2026, Sunday, 11:59 PM)

Assessment

- **Grading Criteria**

Proposal (5 marks)

Component	Excellent (7.50-10.00)	Good (6.00-7.49)	Satisfactory (5.00-5.99)	Weak (4.00-4.99)	Poor (0-3.99)	Component Mark (value from 0.01 to 10.0)
Technical Contents (Weightage: 3)	An innovative and highly feasible solution with exceptional technical depth, showcaseing a thorough research and	A solid and feasible technical solution, indicating good understanding, with a clear	A technically sound solution with basic feasibility, demonstrating an adequate understanding of the	A solution with limited feasibility or technical depth with missing components in the proposal.	The technical solution is unfeasible or irrelevant, with missing components or a poor	3
Structure and language (Weightage: 2)	Well-structured, clear, and concise propsoal. Professional and error-free language, with appropriate technical	Well-organized with clear language. Minor language or grammatical errors, good use of technical terms.	Adequately structured with reasonable clarity. Some language or grammatical errors, adequate use	Poorly organized, lacking clarity. Multiple language or grammatical errors. Ineffective or minimal use of	Poorly organized, difficult to follow. Unclear, misuse of technical terms. Lacks visuals,	2
					Overall Mark (100%)	5

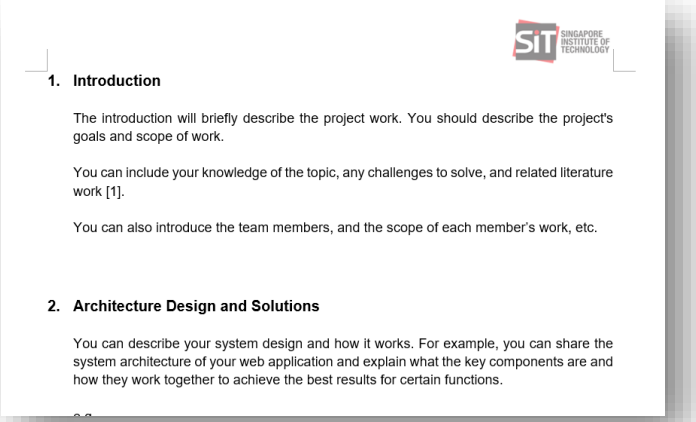
Report and Demo (20 marks)

Component	Excellent (7.50-10.00)	Good (6.00-7.49)	Satisfactory (5.00-5.99)	Weak (4.00-4.99)	Poor (0-3.99)	Component Mark (value from 0.01 to 10.0)
Technical Contents & source codes (Weightage: 10)	Addressing a technical problem with a sound approach, comprehensive background study and in-depth discussion. Complete of codes with comments and	Addressing a technical problem with a structured approach, good background study and logical discussion.	Addressing a technical problem with a structured approach with some background study and general discussion. Workable code.	Technical content is weak with minimum background study and very little discussion. Functions but with some errors in codes.	Technical content is very weak with no background study and no/irrelevant discussion. Not functionable codes.	10
Language and Expression (Weightage: 2)	Clear writing with effective use of simple and complex sentence structures. Excellent paragraphing and text flow.	Clear writing with effective use of simple and complex sentence structures. Good paragraphing and	Clear writing with use of simple and complex sentence structures. Paragraphing and text flow are	Simple sentence structures are used with marginally acceptable paragraphing and text flow. Too	Poorly written report with too many grammatical and spelling mistakes. Poor sentence,	2
Video demo (Weightage: 8)	Exellent, innovative and complete demo of functions and features.	Appropriate, interesting and complete demo of functions and	Good showcase of features, complete demo of functions and	Incomplete demo of functions and features.	Cannot demo basic functions and features.	8
					Overall Mark (100%)	20

Individual Contribution (5 marks)

SN	Name of Team Member	Student ID	Responsibilty	Contribution (%)
1				
2				
3				
4				
5				
6				
			Total	

- **Report Template @ x-site**



1. **Introduction**

The introduction will briefly describe the project work. You should describe the project's goals and scope of work.

You can include your knowledge of the topic, any challenges to solve, and related literature work [1].

You can also introduce the team members, and the scope of each member's work, etc.

2. **Architecture Design and Solutions**

You can describe your system design and how it works. For example, you can share the system architecture of your web application and explain what the key components are and how they work together to achieve the best results for certain functions.

Example of possible topics, but not limited to the following

- Cloud-Based AI-Powered Help Desk System with AI Chatbots
  A web-based help desk platform with an AI Chatbot for users to submit tickets and get responses

- Cloud-Powered E-Learning Platform with Auto-Scaling
  An online e-learning platform with live course materials and online tracking.

- Cloud-Powered AR Shopping Experience
  A web-based AR shopping platform where customers can visualize products in real-world environments using a smartphone.

- Cloud-hosted Web-based Virtual Reality Real Estate Tour
  Real estate virtual home tours, interior design, property sales.

- Cloud-Powered Smart Health Monitoring App
  A mobile health-tracking app that collects real-time vital signs using smartphone sensors or wearables.